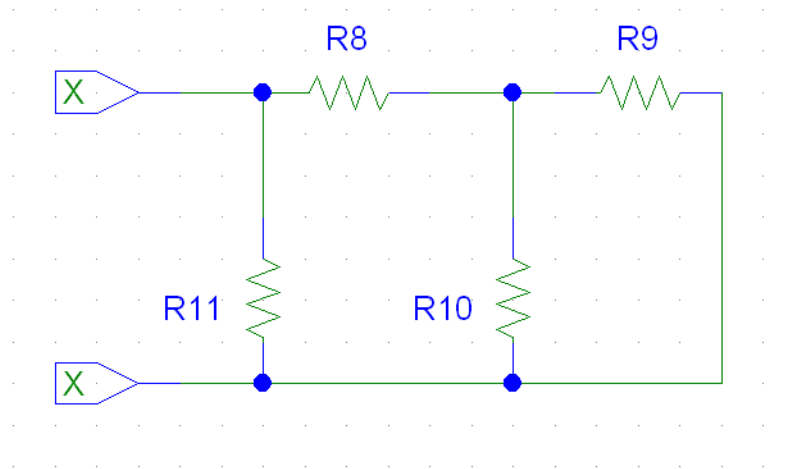




Problem 01



In the circuit, $R_8=45\text{ k}$, $R_9=40\text{ k}$, $R_{10}=30\text{ k}$, $R_{11}=35\text{ k}$;
The voltage source $V_s=16\text{ V}$ is connected to the terminals "X".

1) What is the equivalent resistance between the terminals "X"? (Unit: Ohm)

22389.7059 Ohm

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

2) What is the total power delivered into the circuit by the voltage source V_s ? (Unit: W)

0.0114 W

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

3) What is the power dissipated in the resistor R_{11} ? (Unit: W)

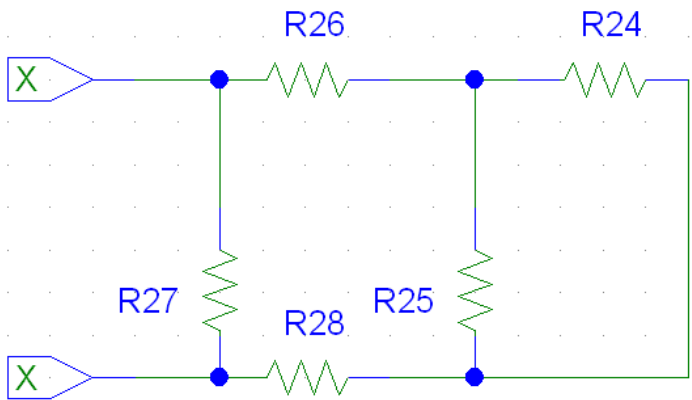
0.0073 W

Computer's answer now shown above. **You are correct.** [Previous Tries](#)



Problem 02





In the circuit, each resistor is 97000 Ohm. Find the equivalent resistance between terminals "X" (unit: Ohm).

69287.1000 Ohm

Computer's answer now shown above. **You are correct.** [Previous Tries](#)



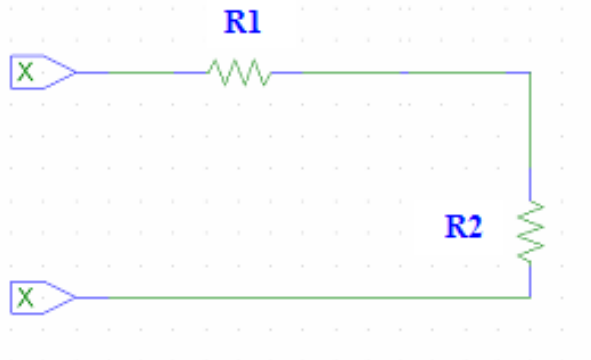
Problem 03



The circuit below is used as a voltage divider. The input voltage is 12 V. The output voltage (that is the voltage across R2) must be 5.1 V.

The current consumed by the divider must be 7.9 mA.

Find the required resistances R1 and R2. (Unit:Ohm)



R1 = 873.4177 Ohm

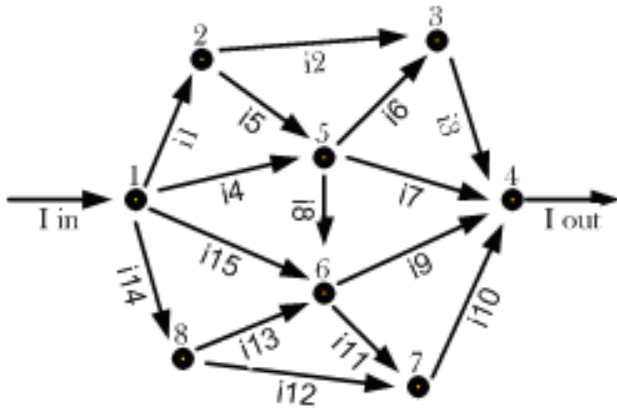
R2 = 645.5696 Ohms

Computer's answer now shown above. **You are correct.** [Previous Tries](#)



Problem 04



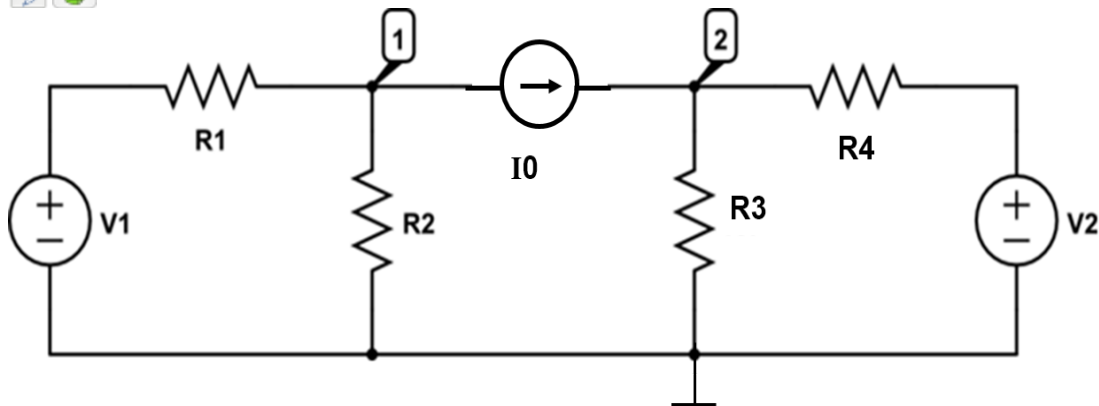


In the figure, if $I_{in} = 3.5$ A, $i_1 = 5.7$ A, $i_2 = 4.6$ A, $i_4 = 9.9$ A, $i_{15} = 5.2$ A, what is the current i_5 ? (UNIT:A)
1.1000 A

Computer's answer now shown above. **You are correct.** [Previous Tries](#)



Problem 05



In the circuit, $V_1 = 120$ V, $R_1 = 8$ Ohm, the current through R_1 flowing from left to right, $i_{R1} = 3$ A.

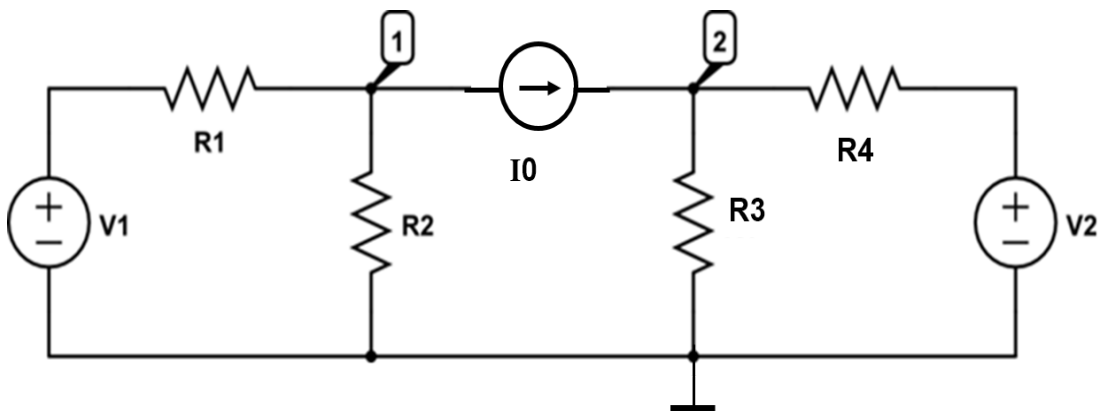
Find the potential of the node "1" (Unit: V)
96 V

Computer's answer now shown above. **You are correct.** [Previous Tries](#)



Problem 06





In the circuit, $V_1=122$ V, $V_2=71$ V, $R_3=17$ Ohm, the current through R_3 flowing downwards, $i_{R3} = 2$ A.

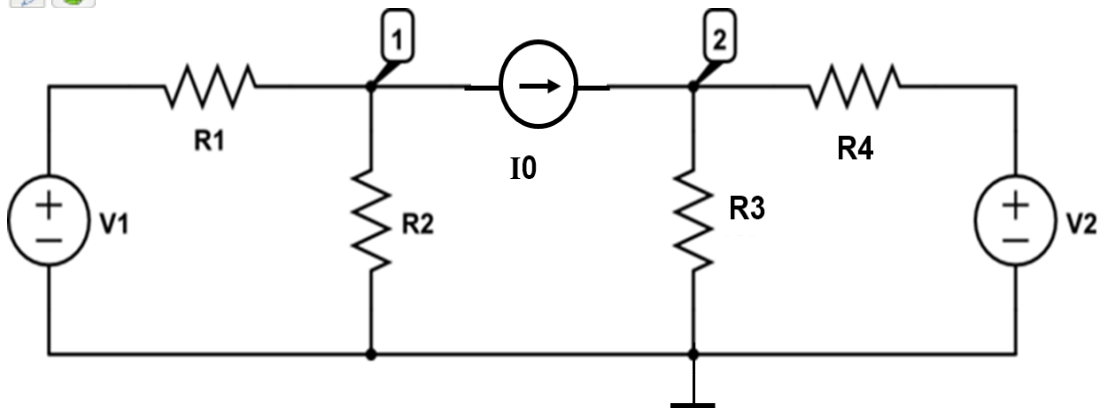
Find the potential of the node "2" (Unit: V)

34 V

Computer's answer now shown above. **You are correct.** [Previous Tries](#)



Problem 07



In the circuit, $V_2=67$ V, $R_4=12$ Ohm, the current through R_4 flowing from left to right, $i_{R4} = 3$ A.

Find the potential of the node "2" (Unit: V)

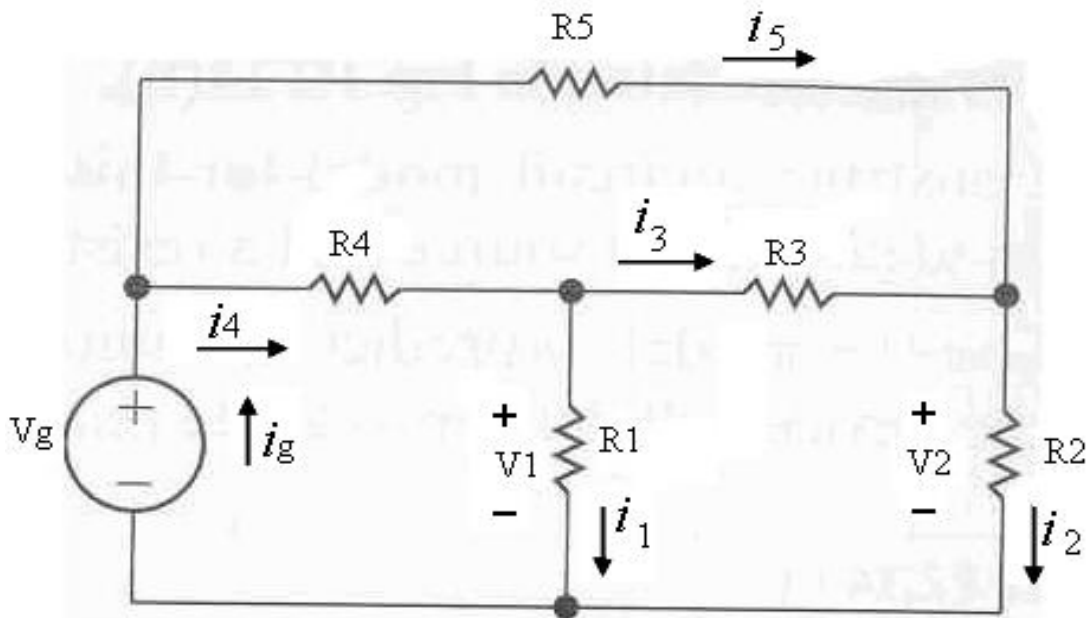
103 V

Computer's answer now shown above. **You are correct.** [Previous Tries](#)



Problem 08





In the circuit, $V_g = 108 \text{ V}$, $R_1 = 11 \text{ Ohm}$, $R_2 = 8 \text{ Ohm}$, $R_3 = 9 \text{ Ohm}$, $R_4 = 4 \text{ Ohm}$ and $R_5 = 13 \text{ Ohm}$. The current $i_5 = 4.28 \text{ A}$.

(A) What is the current i_1 ? (Unit: A)

6.5990 A

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

(B) What is the power delivered by the voltage source V_g into the circuit? (Unit: W)

1418.8020 W

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

(C) What is the power dissipated in the resistor R_2 ? (Unit: W)

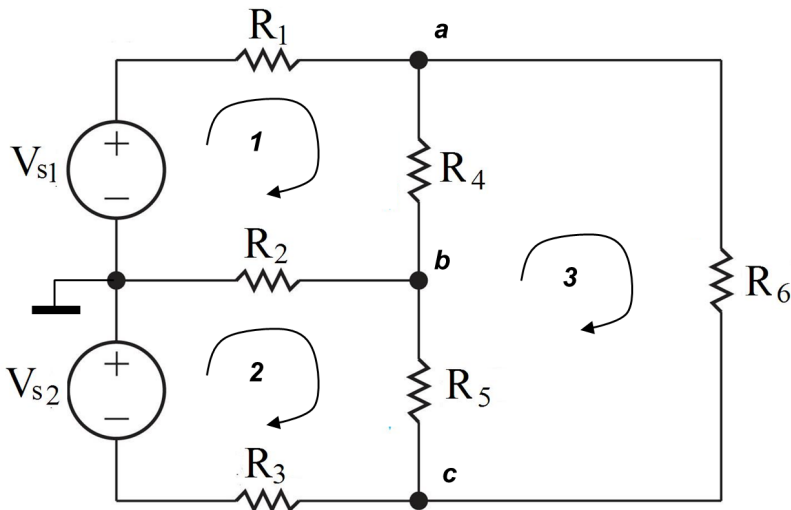
341.9710 W

Computer's answer now shown above. **You are correct.** [Previous Tries](#)



Problem 09





In the circuit, $V_{s1} = 11$ V, $V_{s2} = 21$ V, $R_1 = 3.5$ Ohm, $R_2 = 7$ Ohm, $R_3 = 5.5$ Ohm, $R_4 = 5.5$ Ohm, $R_5 = 4$ Ohm, $R_6 = 5$ Ohm.

A. Assume that the mesh currents for the three meshes shown in the circuit diagram are defined as i_1 , i_2 and i_3 correspondingly.

Using the notations of the caption below circuit diagram (case sensitive!), write the mesh equations for the circuit keeping all the terms on the left-hand side (LHS), so that the equation appears as " LHS = 0"

Enter the left-hand side of mesh equation for the mesh "1"

$$-V_{s1} + i_1 R_1 + (i_1 - i_3) R_4 + (i_1 - i_2) R_2 = 0;$$

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

Enter the left-hand side of mesh equation for the mesh "2"

$$-V_{s2} + (i_2 - i_1) R_2 + (i_2 - i_3) R_5 + i_2 R_3 = 0;$$

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

Enter the left-hand side of mesh equation for the mesh "3"

$$(i_3 - i_2) R_5 + (i_3 - i_1) R_4 + i_3 R_6 = 0;$$

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

B. Using mesh analysis, answer the following questions.

1) Find the current through the resistor R_1 in the direction from left to right. (Unit: A)

$$2.4578 \text{ A}$$

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

2) Find the current through the resistor R_2 in the direction from right to left. (Unit: A)

$$-0.265774262245209 \text{ A}$$

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

3) Find the current through the resistor R_3 in the direction from right to left. (Unit: A)

$$2.7236 \text{ A}$$

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

4) Find the potential of the node "a" with respect to the ground node indicated in the circuit diagram.

(Unit: V)
2.3977 V

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

5) Find the potential of the node "b" with respect to the ground node indicated in the circuit diagram.
(Unit: V)
-1.86041983571646 V

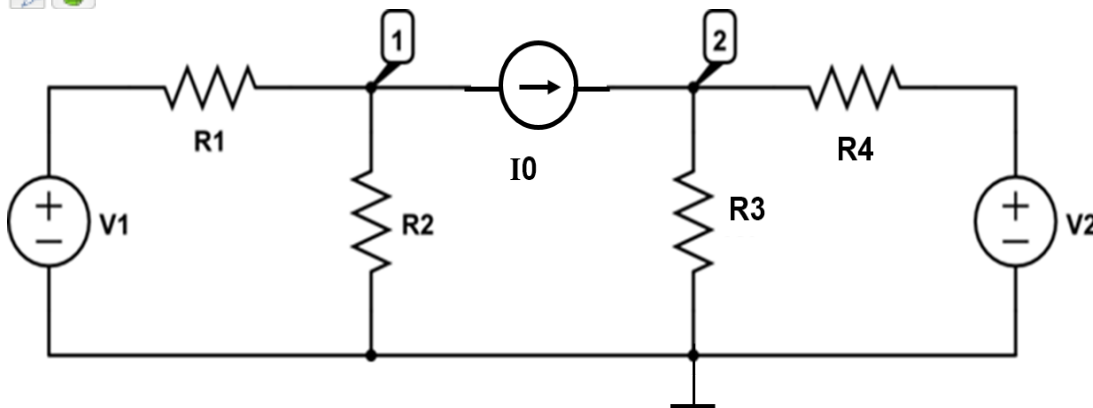
Computer's answer now shown above. **Incorrect.** Tries 3/3 [Previous Tries](#)

6) Find the potential of the node "c" with respect to the ground node indicated in the circuit diagram.
(Unit: V)
-6.02032248250684 V

Computer's answer now shown above. **Incorrect.** Tries 3/3 [Previous Tries](#)



Problem 10



In the circuit, $V_1=129$ V, $V_2=70$ V, $I_0=2$ A, $R_1=7$ Ohm, $R_2=50$ Ohm, $R_3=20$ Ohm, $R_4=23$ Ohm. Solve the circuit using nodal analysis and answer the questions below.

1) Find the current through R_1 flowing in the direction from left to right. (Unit: A)
4.0175 A

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

2) Find the current through R_2 flowing downwards. (Unit: A)
2.0175 A

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

3) Find the current through R_3 flowing downwards. (Unit: A)
2.6977 A

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

4) Find the current through R_4 flowing from left to right. (Unit: A)
-0.697674418604651 A

Computer's answer now shown above. **You are correct.** [Previous Tries](#)

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